

Dense RAG Pipeline

FAISS-Based Dense Retrieval Architecture

RAG-Based Energy Forecasting

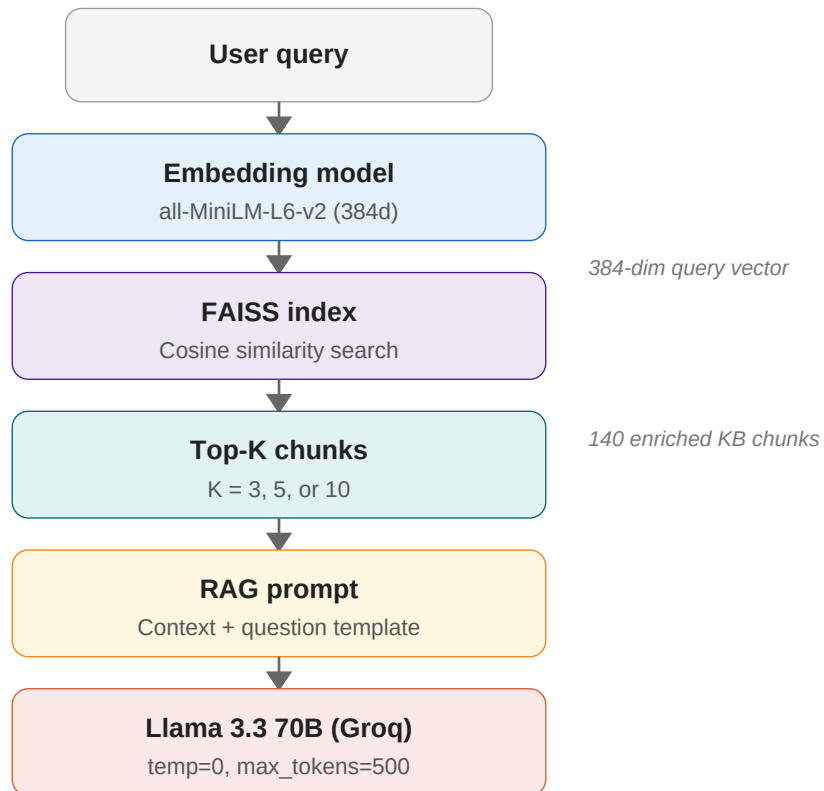
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1. Overview

The Dense RAG pipeline (EXP_02) serves as the primary retrieval baseline. It uses FAISS (Facebook AI Similarity Search) with all-MiniLM-L6-v2 embeddings to perform cosine similarity search over the energy forecasting knowledge base. Retrieved chunks are passed as context to Llama 3.3 70B (via Groq) for answer generation.

2. Architecture Diagram



3. Architecture Components

3.1 Embedding Model

Model: sentence-transformers/all-MiniLM-L6-v2 (384-dimensional embeddings). Both knowledge base chunks and user queries are embedded using the same model to ensure vector space alignment. Embeddings are L2-normalised for cosine similarity.

3.2 FAISS Index

Index type: IndexFlatIP (inner product on normalised vectors = cosine similarity). The index stores all KB chunk embeddings and supports exact nearest-neighbour search. Index files: index.faiss (vectors) + index.pkl (metadata/docstore).

3.3 Retrieval

Given a user query, the DenseRetriever embeds the query and performs similarity_search over the FAISS index to retrieve the top-K most similar chunks. K values tested: 3, 5, 10. Each retrieved chunk includes its row_id, similarity score, and full metadata (dataset, granularity, zone).

3.4 Generation

Retrieved chunks are formatted into a structured context block using format_docs(). The RAG_PROMPT template combines the context with the user question and sends it to Llama 3.3 70B via Groq API (temperature=0, max_tokens=500) for deterministic generation.

4. Pipeline Flow

- Step 1: User query received
- Step 2: Query embedded via all-MiniLM-L6-v2
- Step 3: FAISS similarity_search returns top-K chunks
- Step 4: Chunks formatted into context string
- Step 5: RAG_PROMPT constructed with context + question
- Step 6: Llama 3.3 70B generates grounded answer
- Step 7: Answer evaluated against golden dataset

5. Evaluation Metrics

Metric	Source	Purpose
Recall@K	Custom	Fraction of expected chunks retrieved
Precision@K	Custom	Fraction of retrieved chunks that are relevant
MRR	Custom	Rank position of first relevant chunk
nDCG@K	Custom	Rank-weighted retrieval quality
Faithfulness	RAGAS	Answer grounded in retrieved context
Answer Relevancy	RAGAS	Answer addresses the question
Context Precision	RAGAS	Retrieved context is relevant
Context Recall	RAGAS	Required context was retrieved
Hallucination Rate	Custom	Unsupported claims in answer
Semantic Similarity	Custom	Cosine similarity vs reference answer

6. Key Files

File	Purpose
src/retrieval/dense.py	DenseRetriever class
src/embedding/embedder.py	get_embeddings_model()
src/embedding/faiss_store.py	build/load FAISS index
src/rag/prompts.py	RAG_PROMPT template
experiments/exp_02_dense_rag.py	Experiment orchestration